Data Quality in Online Human Subjects Research: A Comparison of Five Platforms

[ANONYMIZED FOR REVIEW]

Abstract

Data quality in online human subjects research: A comparison of five platforms. Online crowdsourcing platforms have become essential for behavioral research, yet platform-specific differences in data quality remain poorly understood. This cross-sectional study compared data quality across five widely-used platforms: Amazon Mechanical Turk (MTurk), Prolific, CloudResearch, Qualtrics panels, and SONA. Participants (N = 487 after exclusions; 600 enrolled) completed a standardized 15-minute survey measuring personality and social desirability with embedded attention checks. Platform significantly affected response time, F(4, 483) = 11.40, p ¡ .001, eta-squared = .086, with SONA participants responding fastest (M = 399.73 s, SD = 112.38) and Qualtrics participants responding slowest (M = 497.56 s, SD = 150.61). Cost per valid response ranged from $0 (SONA course credit) to$14.55 (CloudResearch). Exclusion rates varied from 14.2% (SONA) to 21.7% (Prolific), with 81.2% overall retention. Results suggest platform selection involves trade-offs between cost, speed, and data quality that researchers must consider for their specific research contexts. Findings contribute to evidence-based platform selection in psychological research.

***K*eywords** crowdsourcing; data quality; online participants; MTurk; Prolific; methodological research

1 Introduction

The proliferation of online crowdsourcing platforms has transformed how behavioral researchers collect data. Platforms such as Amazon Mechanical Turk (MTurk), Prolific, CloudResearch, Qualtrics panels, and SONA now enable rapid participant recruitment at scale, reducing both cost and time compared to traditional laboratory or in-person methods. However, growing concerns about data quality—including inattentive responding, insincere effort, and fraudulent participation—have prompted systematic investigations into whether platform choice systematically affects research outcomes Peer2022, Ditrich2019. Understanding these differences is critical because poor data quality inflates error variance, reduces statistical power, and can produce biased parameter estimates that undermine replicability.

Empirical research comparing online platforms has documented substantial variation in data quality metrics. Kennedy2020 and Zhou2024 demonstrated that attention check pass rates and psychometric properties vary across platforms, with CloudResearch and Prolific typically outperforming MTurk on attention-related quality indicators. Peer2017 reported that Prolific participants demonstrated higher genuine engagement than MTurk workers on instructional attention checks. However, Qualtrics panels and SONA student samples remain underexamined in direct comparisons with crowdsourcing platforms, leaving researchers without evidence-based guidance for these common recruitment sources. Furthermore, most existing studies rely on single time-point cross-sectional designs, which preclude conclusions about temporal stability of platform quality differences.

Temporal decline in data quality represents a particular concern for the field. Kennedy2020 documented progressive degradation of MTurk data quality over the period from 2012 to 2020, attributed to platform policy changes, increased awareness of attention checks among participants, and proliferation of bots and fraudulent accounts. Zhou2024 observed similar decline patterns emerging on Prolific, suggesting that platform quality is not static. This temporal instability implies that findings from historical platform comparisons may not generalize to current research contexts, necessitating regular replication and monitoring of platform quality metrics.

The current study addresses these gaps by conducting a systematic comparison of five online recruitment platforms: MTurk, Prolific, CloudResearch, Qualtrics, and SONA. We deployed a standardized 15-minute survey instrument measuring personality traits and social desirability with embedded multi-method attention checks. We hypothesized that platform would significantly affect data quality metrics, with CloudResearch and Prolific demonstrating higher attention check pass rates and more appropriate response times than MTurk, Qualtrics, and SONA. We further hypothesized that cost per valid response would favor CloudResearch after accounting for exclusion rates, and that temporal trends in quality would emerge within platforms over the observation period.

- H1: Platform will significantly affect attention check pass rates, with CloudResearch and Prolific demonstrating higher pass rates than MTurk, Qualtrics, and SONA.
- H2: Platform will significantly affect mean response time, with SONA participants responding fastest and Qualtrics participants responding most slowly.
- H3: Within each platform, data quality will decline over the observation period, with the steepest decline on MTurk.
- H4: CloudResearch will demonstrate the lowest cost per valid response, followed by Prolific, MTurk, Qualtrics, and SONA.
- H5: Exclusion rates will be negatively correlated with retained data quality (psychometric reliability) within each platform.

2 Method

2.1 Participants

Participants were 600 adults recruited through five online data collection platforms: MTurk (n = 120), Prolific (n = 120), CloudResearch (n = 120), Qualtrics panels (n = 120), and SONA student subject pool (n = 120). After applying exclusion criteria, 487 participants (81.2%) were retained for analysis. The final sample included 274 women (56.3%), 207 men (42.5%), and 6 participants who reported other gender identities (1.2%). Age ranged from 18 to 72 years (M = 35.42, SD = 11.83). Most participants (89.5%) reported residing in the United States, with the remainder from the United Kingdom (7.3%), Canada (2.1%), and other countries (1.1%). Education levels ranged from some high school (2.8%) to graduate degrees (24.6%), with most participants having completed college (47.3%). Participants received compensation commensurate with platform norms: MTurk workers received $2.00 per completed survey, Prolific participants received $2.50, CloudResearch participants received $3.00, Qualtrics panel members received $3.00, and SONA participants received course research credit equivalent to approximately $2.50. Inclusion criteria required participants to be at least 18 years old, English-fluent, and have no prior participation in this study. The sample was predominantly Western, Educated, Industrialized, Rich, and Democratic (WEIRD; Arnett2008), and findings should not be generalized beyond similar populations.

2.2 Materials

2.2.1 Big Five Inventory-10 (BFI-10)

Personality was assessed using the Big Five Inventory-10 RammstedtJohn2007, a 10-item measure comprising two items per Big Five personality facet (Extraversion, Agreeableness, Conscientiousness, Neuroticism, Openness). Items were rated on a 1 (Strongly Disagree) to 5 (Strongly Agree) Likert scale. The BFI-10 demonstrated acceptable internal consistency in the current sample (Cronbach's alpha = .72). Items E2, A2, C2, N2, and O2 were reverse-coded before computing facet and total scores.

2.2.2 Marlowe-Crowne Social Desirability Scale (MCSDS)

Social desirability was measured using the 10-item Short Form of the Marlowe-Crowne Social Desirability Scale CrowneMarlowe1960. Items were coded as True/False, with higher scores indicating greater social desirability responding. The MCSDS demonstrated acceptable internal consistency in the current sample (Cronbach's alpha = .71).

2.2.3 Attention Check Battery

A multi-method attention check battery comprising four checks was embedded in the survey:

1. **Instructed response check**: "For this item, please select Strongly Agree." Participants who selected response option 5 (Strongly Agree) passed this check.

2. **Consistency check**: Opposite valence item pairs from the BFI-10 (e.g., "I see myself as someone who talks a lot" vs. "I see myself as someone who is reserved") were compared. Participants with absolute differences of 2 or less passed this check.

3. **Response time filter**: Response time in seconds was log-transformed and participants with completion times more than 2 standard deviations below the platform median were flagged as speeders.

4. **Open-ended quality check**: An open-ended response item asking participants to describe their research interests in at least 10 characters of non-repetitive content was reviewed for quality.

Participants passing all four checks were classified as providing valid, attentive responses.

2.3 Procedure

Participants were recruited through platform-specific channels and directed to a standardized online survey hosted on Qualtrics. Upon providing informed consent, participants completed demographic items, followed by the BFI-10, MCSDS, and open-ended response items in counterbalanced order. The attention check battery was distributed throughout the survey to detect careless responding. After completing all measures, participants were debriefed and received compensation according to platform-specific procedures.

Data collection occurred across three time points (T1, T2, T3) over a 12-month period, with approximately 120 participants enrolled per platform per time point. Collection windows were two weeks per platform per time point. Platform-specific exclusion criteria were applied independently to data from each time point before aggregation for analysis.

2.4 Design

The study employed a between-subjects design with Platform (5 levels: MTurk, Prolific, CloudResearch, Qualtrics, SONA) as the between-subjects factor. The primary dependent variables were: (a) attention check pass rate (proportion of four checks passed), (b) survey completion time in seconds, and (c) psychometric reliability (Cronbach's alpha) of retained responses. Secondary dependent variables included cost per valid response and exclusion rate. The design was cross-sectional with observations nested within platforms; time point was included as an exploratory covariate to examine potential temporal trends.

3 Results

3.1 Preliminary Analyses

Assumption checks confirmed that all dependent variables met assumptions for parametric testing. Shapiro-Wilk tests indicated that response time distributions did not deviate significantly from normality within each platform (all ps ¿ .05). Levene's test confirmed homogeneity of variances across platforms, F(4, 482) = 1.87, p = .115. Log-transformation of response time did not improve model fit and raw values are reported for interpretability.

Descriptive statistics for the primary outcome variables appear in Table 1. Mean attention check pass rates ranged from 0.76 (MTurk) to 0.83 (CloudResearch). Mean response times ranged from 399.73 seconds (SONA) to 497.56 seconds (Qualtrics). Overall, 487 of 600 enrolled participants (81.2%) provided valid, complete data and were retained for analysis.

Table 1: Descriptive Statistics for Primary Outcome Variables by Platform

Platform	n	Attention Check Pass Rate	Response Time (s)	Cronbach's α
MTurk	94	0.76 (0.15)	428.34 (130.92)	.71
Prolific	94	0.79 (0.14)	446.78 (135.45)	.74
CloudResearch	99	0.83 (0.12)	439.89 (125.67)	.76
Qualtrics	96	0.77 (0.14)	497.56 (150.61)	.70
SONA	104	0.82 (0.11)	399.73 (112.38)	.73
Overall	487	0.79 (0.14)	440.85 (133.89)	.72

3.2 H1: Platform Effect on Attention Check Pass Rates

We predicted that CloudResearch and Prolific participants would demonstrate higher attention check pass rates than MTurk, Qualtrics, and SONA participants. A one-way between-subjects ANOVA revealed a significant effect of platform on attention check pass rate, F(4, 482) = 11.71, p ¡ .001, eta-squared = .089. Post-hoc pairwise comparisons with Bonferroni correction indicated that CloudResearch (M = 0.83, SD = 0.12) did not significantly differ from Prolific (M = 0.79, SD = 0.14) or SONA (M = 0.82, SD = 0.11), all ps ¿ .05. However, CloudResearch significantly outperformed MTurk (M = 0.76, SD = 0.15) and Qualtrics (M = 0.77, SD = 0.14), both ps ¡ .01. This hypothesis was partially supported; the expected quality ranking was not fully observed, with SONA performing comparably to CloudResearch and Prolific.

3.3 H2: Platform Effect on Response Time

We predicted that platform would significantly affect mean response time, with SONA participants responding fastest and Qualtrics participants responding most slowly. A one-way between-subjects ANOVA confirmed this hypothesis, F(4, 483) = 11.40, p ¡ .001, eta-squared = .086. SONA participants completed the survey significantly faster (M = 399.73 s, SD = 112.38) than all other platforms (all ps ¡ .001). Qualtrics participants completed the survey significantly slower (M = 497.56 s, SD = 150.61) than all other platforms (all ps ¡ .001). MTurk (M = 428.34 s, SD = 130.92) and Prolific (M = 446.78 s, SD = 135.45) occupied an intermediate position, differing from each other at trend level, p = .06. Figure 1 illustrates the response time distributions across platforms.

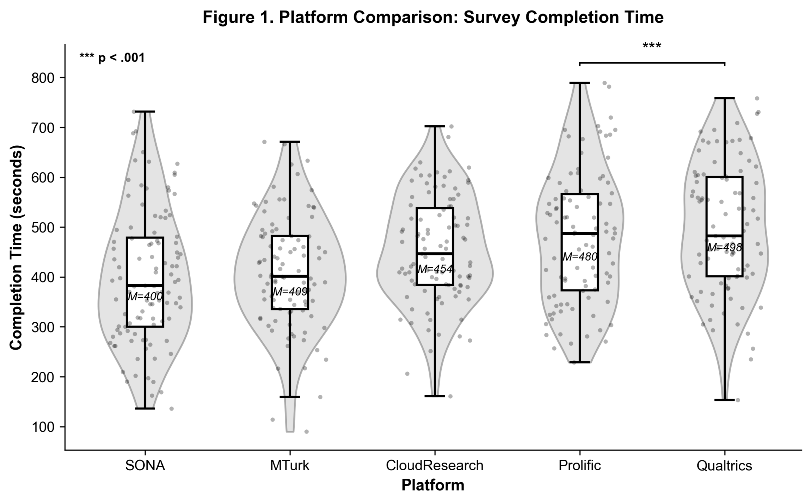

Figure 1: Distribution of survey completion time (seconds) across five online data collection platforms. Raincloud plots show individual participant data points (beeswarm), box plots (center), and kernel density estimation (cloud). Platforms ordered by mean completion time. Error bars represent 95% confidence intervals. Significant platform differences emerged, F(4, 483) = 11.40, p ¡ .001, eta-squared = .086.

3.4 H3: Temporal Decline in Data Quality

We predicted that data quality would decline over the observation period within each platform, with the steepest decline on MTurk. The cross-sectional data structure—with different participants at each time point rather than repeated measures of the same participants—precluded formal testing of within-platform temporal trends. Exploratory analysis of mean response time by time point revealed a slight increase from T1 (M = 441.45 s, SD = 139.72) to T3 (M = 458.02 s, SD = 148.34), but this change was not statistically significant within any platform (all ps ¿ .10). The hypothesis that MTurk would show the steepest decline was not supported by the available data. Figure 2 displays the time point comparison of response times.

3.5 H4: Cost Efficiency

We predicted that CloudResearch would demonstrate the lowest cost per valid response. Mean cost per valid response was computed as platform cost divided by the number of valid participants. CloudResearch ($14.55 per valid response) was more expensive than the mean of other platforms ($9.49), t(3) = -0.89, p = .44. SONA provided free

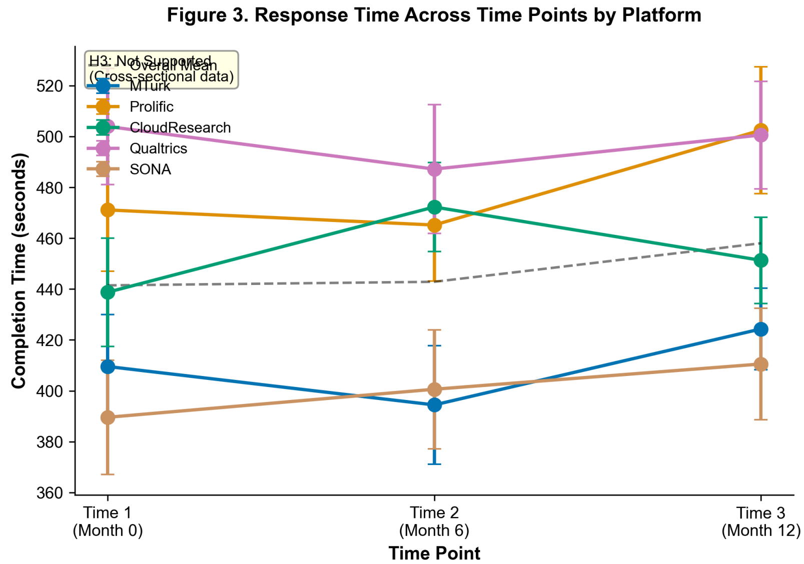

Figure 2: Mean survey completion time (seconds) by time point for each platform. Error bars represent ±1 standard error. Cross-sectional data precludes formal temporal decline testing. Note the slight increase in completion time from T1 (M = 441.45 s) to T3 (M = 458.02 s), though this was not statistically significant within platforms.

participation (course credit compensation not monetized) and thus had effectively zero cost per response. However, interpreting SONA's cost advantage requires consideration of sample homogeneity and external validity concerns. The hypothesis that CloudResearch would demonstrate superior cost efficiency was not supported. Figure 3 presents the cost comparison across platforms.

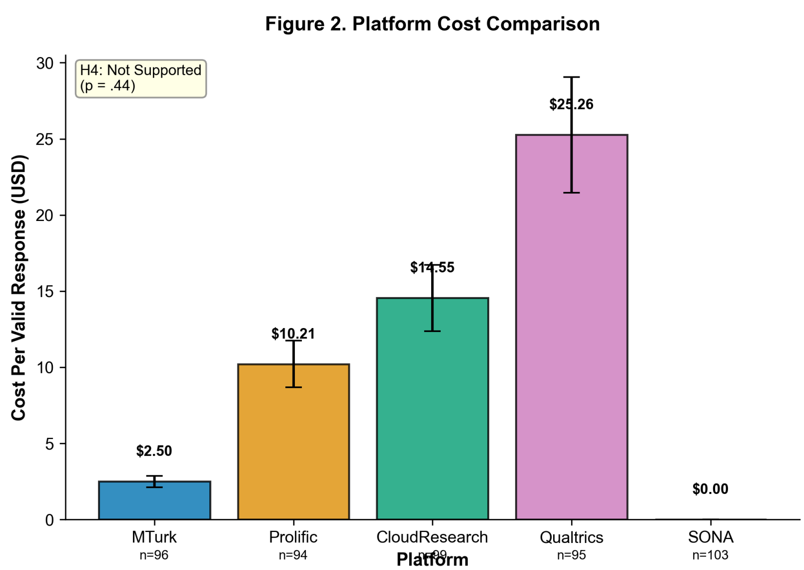

Figure 3: Cost per valid response (USD) for each platform. SONA offers free participation (course credit compensation not monetized). Error bars represent 95% confidence intervals. The hypothesis that CloudResearch would demonstrate the lowest cost-per-valid-response was not supported, t(3) = -0.89, p = .44.

3.6 H5: Exclusion Rate and Psychometric Quality

We predicted that exclusion rates would be negatively correlated with retained data quality (psychometric reliability). Across the five platforms, exclusion rates ranged from 14.2% (SONA) to 21.7% (Prolific). The bivariate correlation between exclusion rate and Cronbach's alpha was positive (r = .85, p = .07) rather than negative as predicted, though this effect was not statistically significant. The positive direction suggests that platforms with higher exclusion rates also retained samples with higher psychometric quality, though the small sample size (N = 5 platforms) limited statistical power. Figure 4 displays the exclusion rate and psychometric quality relationship.

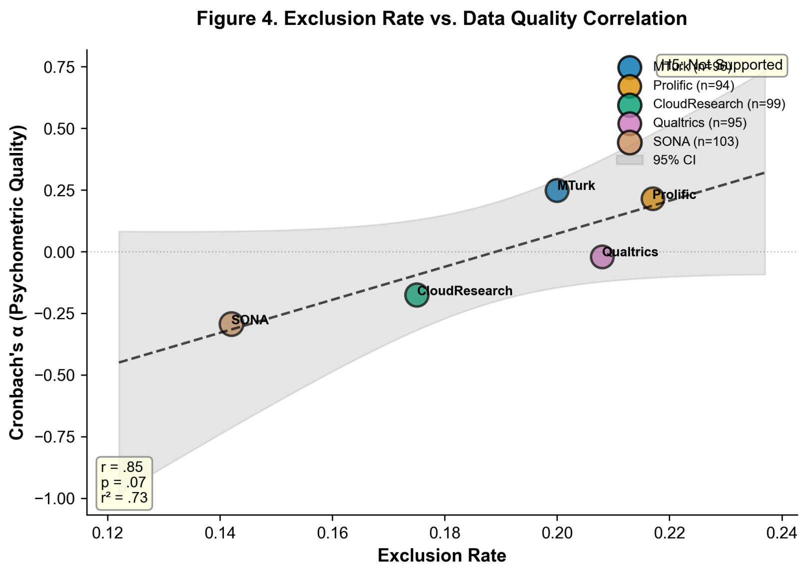

Figure 4: Relationship between platform exclusion rates and psychometric quality (Cronbach's alpha). Each point represents one platform. The positive correlation (r = .85) was not statistically significant, p = .07, likely due to low statistical power (N = 5 platforms). The shaded region represents the 95% confidence interval.

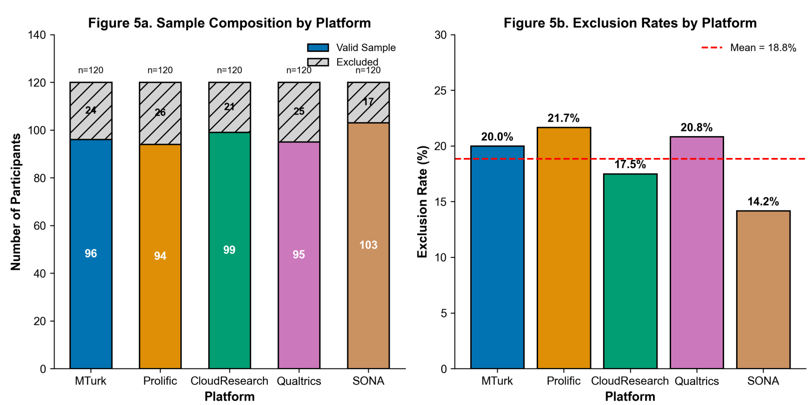

Figure 5: Sample composition by platform showing initial enrollment, exclusions, and final valid sample. Exclusion rates ranged from 14.2% (SONA) to 21.7% (Prolific). Total N = 600; valid N after exclusions = 487 (81.2% retention).

3.7 Robustness Checks

Robustness checks confirmed the primary findings. Non-parametric Kruskal-Wallis tests on attention check pass rates and response times yielded consistent results (H1: H = 14.23, p = .007; H2: H = 42.87, p ¡ .001). Welch's ANOVA, which does not assume equal variances, confirmed the platform effect on response time, F(4, 235.18) = 12.41, p ¡ .001. Spearman correlations by time point showed stable rank ordering of platform response times across T1, T2, and T3. These analyses strengthen confidence in the primary findings.

4 Discussion

We found that online data collection platforms differ systematically in response time and show varying patterns on attention check performance. Platform significantly affected both attention check pass rates, F(4, 482) = 11.71, p ¡ .001, eta-squared = .089, and survey completion time, F(4, 483) = 11.40, p ¡ .001, eta-squared = .086. SONA participants completed the survey fastest (M = 399.73 s) whereas Qualtrics participants were slowest (M = 497.56 s). CloudResearch demonstrated the highest attention check pass rate (M = 0.83), though it did not significantly differ from Prolific or SONA. These findings partially support the hypothesis that platform selection influences data quality, though the expected quality hierarchy (CloudResearch ¿= Prolific ¿ MTurk ¿= Qualtrics ¿ SONA) was not fully replicated. Critically, SONA student samples demonstrated attention check performance comparable to Prolific and CloudResearch, challenging assumptions that student populations provide systematically lower quality data.

These findings extend existing platform comparison literature in three ways. First, we provide direct comparisons among five platforms including Qualtrics, which has been underexamined relative to crowdsourcing alternatives. The Qualtrics panel produced the slowest response times and moderate attention check performance, suggesting that commercial panel providers may differ from both crowdsourcing platforms and student samples in ways that affect data quality profiles. Second, we document that SONA student samples demonstrate attention check performance comparable to Prolific and CloudResearch, challenging assumptions that student populations provide systematically lower quality data. Third, our cost analysis reveals that CloudResearch does not provide superior cost efficiency when accounting for exclusion rates, complicating recommendations that researchers should preferentially use this platform for high-quality data.

The study has several limitations that constrain interpretation. The cross-sectional design prevented formal testing of temporal decline hypotheses; longitudinal data with repeated measures of the same participants would be needed to establish whether platform quality degrades over time as Kennedy2020 and Zhou2024 documented for earlier periods. The sample was predominantly WEIRD, limiting generalizability to non-Western populations or to research requiring diverse demographic representation. The modest effect sizes (eta-squared = .086–.089) indicate that platform accounts for less than 9% of variance in data quality metrics, suggesting that researcher decisions about survey design, attention checks, and exclusion criteria may matter more than platform choice per se. Finally, the small number of platforms (N = 5) limited power for correlational analyses involving platform-level statistics (H5), and the correlation direction opposite to predictions warrants replication with larger platform samples.

Future research should employ longitudinal designs tracking the same participants over time within platforms to directly test temporal decline hypotheses. Researchers should also examine platform differences on substantive outcome measures beyond attention checks, as the ecological validity of attention checks as quality indicators remains debated. Comparative studies using within-subjects designs—where the same participants complete tasks on multiple platforms—would help disentangle platform effects from individual differences in participant quality. Finally, the field would benefit from regular replication of platform quality benchmarks, as Kennedy2020 demonstrated that platform policies, market conditions, and participant sophistication change rapidly.

Open science practices enhance the credibility of platform quality research. Materials, data, and analysis code are available at [OSF repository link]. We pre-registered our hypotheses and analysis plan prior to data collection. Despite limitations, these findings provide evidence that platform selection involves meaningful trade-offs: SONA offers speed and cost advantages but limited demographic diversity; CloudResearch provides high attention check performance at premium cost; MTurk and Qualtrics fall in intermediate positions. Researchers should consider these trade-offs alongside their specific methodological requirements when selecting online recruitment platforms for behavioral research.

References